Scheme of MAT-2122.

11. LU Decomposition (04)

$x+y+z=1,\ 3x+y-3z=5,\ x-2y-5z=10$

$$A=\begin{bmatrix}1&1&1\\3&1&-3\\1&-2&-5\end{bmatrix}=\begin{bmatrix}1&0&0\\l_{21}&1&0\\l_{31}&l_{32}&1\end{bmatrix}\begin{bmatrix}u_{11}&u_{12}&u_{13}\\0&u_{22}&u_{23}\\0&0&u_{33}\end{bmatrix}$$

$$\begin{bmatrix}1&1&1\\3&1&-3\\1&-2&-5\end{bmatrix}\sim\begin{bmatrix}1&1&1\\0&-2&-6\\0&-3&-6\end{bmatrix}\sim\begin{bmatrix}1&1&1\\0&-2&-6\\0&0&3\end{bmatrix}\longrightarrow (1m)$$

$R_2=R_2-3R_1 \qquad R_3=R_3-\frac{3}{2}R_2$

$R_3=R_3-R_1$

$A=LU$

$$\begin{bmatrix}1&1&1\\3&1&-3\\1&-2&-5\end{bmatrix}=\begin{bmatrix}1&0&0\\3&1&0\\1&3/2&1\end{bmatrix}\begin{bmatrix}1&1&1\\0&-2&-6\\0&0&3\end{bmatrix}\longrightarrow (1m)$$

$\therefore AX=B \Rightarrow LUX=B$

$LY=B \Rightarrow$

$$\begin{bmatrix}1&0&0\\3&1&0\\1&3/2&1\end{bmatrix}\begin{bmatrix}y_1\\y_2\\y_3\end{bmatrix}=\begin{bmatrix}1\\5\\10\end{bmatrix}$$

$y_1=1$

$\Rightarrow 3y_1+y_2=5 \Rightarrow y_2=2$

$y_1+\frac{3}{2}y_2+y_3=10 \Rightarrow y_3=6$

$\longrightarrow (1m)$

$UX=Y$

$$\begin{bmatrix}1&1&1\\0&-2&-6\\0&0&3\end{bmatrix}\begin{bmatrix}x\\y\\z\end{bmatrix}=\begin{bmatrix}1\\2\\6\end{bmatrix}$$

$3z=6 \Rightarrow z=2$

$-2y-6z=2 \Rightarrow y=-7$

$x+y+z=1 \Rightarrow x=6$

$\therefore x=6,\ y=-7,\ z=2 \longrightarrow (1m)$

$$A = \begin{bmatrix} 4 & 3 \\ 2 & 9 \end{bmatrix}$$

(03)

$\lambda^2 - 13\lambda + 30 = 0 \Rightarrow \lambda = \underline{10, 3}$ → (½ M)

$\underline{\lambda = 10}$,

$$\begin{bmatrix} -6 & 3 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$\Rightarrow R_2 = 6R_2 + 2R_1 \Rightarrow \begin{bmatrix} -6 & 3 \\ 0 & 0 \end{bmatrix}$

By taking $y = k$, $x = k/2$ $\therefore \begin{bmatrix} x \\ y \end{bmatrix} = \underline{\begin{bmatrix} k/2 \\ k \end{bmatrix}}$

If, $k = 2$ $\begin{bmatrix} x \\ y \end{bmatrix} = \underline{\begin{bmatrix} 1 \\ 2 \end{bmatrix}}$

$\underline{\lambda = 3}$

$$\begin{bmatrix} 1 & 3 \\ 2 & 6 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

By applying $R_2 = R_2 - 2R_1 \Rightarrow \begin{bmatrix} 1 & 3 \\ 0 & 0 \end{bmatrix}$

By taking $y = k$, $x = -3k$ $\therefore \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -3k \\ k \end{bmatrix}$

If $k = 1$, $\begin{bmatrix} x \\ y \end{bmatrix} = \underline{\begin{bmatrix} -3 \\ 1 \end{bmatrix}}$

(01 M)

$\therefore P = \begin{bmatrix} 1 & -3 \\ 2 & 1 \end{bmatrix}$ then $D = \begin{bmatrix} 10 & 0 \\ 0 & 3 \end{bmatrix}$ & $P^{-1} = \begin{bmatrix} 1/7 & 3/7 \\ -2/7 & 1/7 \end{bmatrix}$ → (½ M)

$$A^3 = PD^3P^{-1} = \begin{bmatrix} 1 & -3 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 10^3 & 0 \\ 0 & 3^3 \end{bmatrix} \begin{bmatrix} 1/7 & 3/7 \\ -2/7 & 1/7 \end{bmatrix} = \underline{\begin{bmatrix} 166 & 417 \\ 278 & 861 \end{bmatrix}}$$ → (01 M)

x	y	$\theta = \pi x/6$	$y\cos\theta$	$y\sin\theta$
0	9	0	9	0
2	18.2	$\pi/3$	9.1	15.7616
4	24.4	$2\pi/3$	−12.2	21.1310
6	27.8	π	−27.8	0
8	27.5	$4\pi/3$	−13.75	−23.8156
10	22	$5\pi/3$	11	−19.0525
	128.9		−24.65	−5.9755

$(0, 2l) = (0, 12)$

$l = 6$

(03)

(01½ M)

$$a_0 = \frac{2}{N}\Sigma y = \frac{2}{6}(128.9) = 42.9667$$

$$a_1 = \frac{2}{N}\Sigma y\cos\theta = \frac{-24.65}{3} = -8.2166$$

$$b_1 = \frac{2}{N}\Sigma y\sin\theta = \frac{-5.9755}{3} = -1.9918$$

(01 M)

$$\therefore f(x) = 21.4833 - 8.2166\cos\theta - 1.9918\sin\theta$$ → (01½ M)

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14. $f(x) = \begin{cases} 1, & |x| < a \\ 0, & |x| > a \end{cases}$ (03)

$$F\{f(x)\} = \frac{1}{\sqrt{2\pi}}\int_{-a}^{a} 1\cdot e^{isx}\,dx = \frac{1}{\sqrt{2\pi}}\left(\frac{e^{isx}}{is}\right)\Big|_{-a}^{a}$$

$$= \frac{1}{\sqrt{2\pi}}\frac{1}{is}(e^{isa} - e^{-isa}) = \sqrt{\frac{2}{\pi}}\cdot\frac{\sin sa}{s}$$ → (01 M)

$$F^{-1}\{F(s)\} = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty}\sqrt{\frac{2}{\pi}}\cdot\frac{\sin sa}{s}e^{-isx}\,ds$$ (01 M)

$$f(x) = \frac{1}{\pi}\int_{-\infty}^{\infty}\frac{\sin sa}{s}e^{-isx}\,ds.$$

By substituting $x = 0$,

$$f(0) = \frac{1}{\pi}\int_{-\infty}^{\infty}\frac{\sin sa}{s}\,ds \Rightarrow 1 = \frac{2}{\pi}\int_{0}^{\infty}\frac{\sin sa}{s}\,ds$$ (01 M)

By substituting $t = sa$.

$$\int_0^\infty \frac{\sin t}{t}\,dt = \frac{\pi}{2}.$$

15. $A = \begin{bmatrix} 3 & 1 \\ 4 & 2 \end{bmatrix}$ QR Decomposition (03)

$a_1 = (3,4) \quad a_2 = (1,2)$.

$v_1 = a_1 = (3,4) \Rightarrow u_1 = \dfrac{(3,4)}{\sqrt{16+9}} = \underline{(3/5, 4/5)}$ → (01M)

$v_2 = a_2 - (a_2 \cdot u_1)u_1$

$= (1,2) - \left((1,2)\cdot(3/5, 4/5)\right)(3/5, 4/5)$

$= (1,2) - \left(3/5 + 8/5\right)(3/5, 4/5) = \underline{(-8/25, \tfrac{6}{25})}$

$u_2 = \dfrac{(-8/25, 6/25)}{\sqrt{\frac{64}{(25)^2} + \frac{36}{(25)^2}}} = \dfrac{25}{10}(-8/25, 6/25) = \underline{(-4/5, 3/5)}$ → (01M)

$\therefore Q = \begin{bmatrix} 3/5 & -4/5 \\ 4/5 & 3/5 \end{bmatrix}$

$R = \begin{bmatrix} a_1 \cdot u_1 & a_2 \cdot u_1 \\ 0 & a_2 \cdot u_2 \end{bmatrix} = \begin{bmatrix} 5 & 11/5 \\ 0 & 2/5 \end{bmatrix}$ → (01M)

16. $f(x) = \begin{cases} 0, & -5 < x < 0 \\ 3, & 0 < x < 5 \end{cases}$, $f(x+10) = f(x)$ (03)

$a_0 = \dfrac{1}{5}\int_0^5 3 \cdot dx = \dfrac{3}{5} \cdot x \Big|_0^5 = \underline{3}$ → (01/2 M)

$a_n = \dfrac{1}{5}\int_0^5 3 \cdot \cos\dfrac{n\pi}{5} x \, dx = \dfrac{3}{5}\left(\dfrac{\sin n\pi/5\, x}{n\pi/5}\right)\Big|_0^5 = \underline{0}$ → (1/2 M)

$b_n = \dfrac{1}{5}\int_0^5 3 \sin\dfrac{n\pi}{5} x \, dx = \dfrac{3}{5}\left(\dfrac{-\cos n\pi/5\, x}{n\pi/5}\right)\Big|_0^5 = \dfrac{-3}{n\pi}(\cos n\pi - \cos 0)$

$= \dfrac{-3}{n\pi}\left((-1)^n - 1\right) = \underline{\dfrac{3(1-(-1)^n)}{n\pi}}$ → (01M)

$\therefore f(x) = \sum_{n=1}^{\infty} \dfrac{3(1-(-1)^n)}{n\pi} \sin n\pi/5\, x + 3/2$ → (01M)

Cholesky Decomposition of $\begin{bmatrix} 9 & 6 & 12 \\ 6 & 13 & 11 \\ 12 & 11 & 26 \end{bmatrix}$ (02)

$A = LL^T$

$$\begin{bmatrix} 9 & 6 & 12 \\ 6 & 13 & 11 \\ 12 & 11 & 26 \end{bmatrix} = \begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix} \begin{bmatrix} l_{11} & l_{21} & l_{31} \\ 0 & l_{22} & l_{32} \\ 0 & 0 & l_{33} \end{bmatrix}$$

$$\begin{bmatrix} l_{11}^2 & l_{11}l_{21} & l_{11}l_{31} \\ l_{11}l_{21} & l_{21}^2+l_{22}^2 & l_{21}l_{31}+l_{22}l_{32} \\ l_{11}l_{31} & l_{21}l_{31}+l_{22}l_{32} & l_{31}^2+l_{32}^2+l_{33}^2 \end{bmatrix} = \begin{bmatrix} 9 & 6 & 12 \\ 6 & 13 & 11 \\ 12 & 11 & 26 \end{bmatrix} \rightarrow 0\frac{1}{2}M$$

$l_{11}^2 = 9 \Rightarrow l_{11} = \underline{3}$

$l_{11}l_{21} = 6 \Rightarrow l_{21} = \underline{2}$

$l_{11}l_{31} = 12 \Rightarrow l_{31} = \underline{4}$

$l_{21}^2 + l_{22}^2 = 13 \Rightarrow l_{22}^2 = 9 \Rightarrow l_{22} = \underline{3}$

$l_{21}l_{31} + l_{22}l_{32} = 11 \Rightarrow l_{32} = \underline{1}$.

$l_{31}^2 + l_{32}^2 + l_{33}^2 = 26 \Rightarrow l_{33}^2 = 9 \Rightarrow l_{33} = \underline{3}$.

} $\rightarrow$ 01 M

$$\therefore \begin{bmatrix} 9 & 6 & 12 \\ 6 & 13 & 11 \\ 12 & 11 & 26 \end{bmatrix} = \begin{bmatrix} 3 & 0 & 0 \\ 2 & 3 & 0 \\ 4 & 1 & 3 \end{bmatrix} \begin{bmatrix} 3 & 2 & 4 \\ 0 & 3 & 1 \\ 0 & 0 & 3 \end{bmatrix} \rightarrow 0\frac{1}{2}M$$

18. $f(x) = \begin{cases} x, & 0 < x < 4 \\ 0, & \text{else} \end{cases}$ (02)

$$F_c\{f(x)\} = \sqrt{\frac{2}{\pi}} \int_0^4 x \cdot \cos sx \, dx = \sqrt{\frac{2}{\pi}} \left\{ x\left(\frac{\sin sx}{s}\right)\Big|_0^4 - \left(\frac{-\cos sx}{s^2}\right)\Big|_0^4 \right\}$$

$\rightarrow$ 01 M

$$= \sqrt{\frac{2}{\pi}} \left\{ 4\frac{\sin 4s}{s} + \frac{\cos 4s}{s^2} - \frac{1}{s^2} \right\} \rightarrow 01\,M$$

7. Half-range sine series of $f(x) = c - x$, $0 < x < c$

$$f(x) = \sum_{n=1}^{\infty} b_n \sin \frac{n\pi}{c} x.$$ (02)

$$b_n = \frac{2}{c} \int_0^c (c-x) \sin \frac{n\pi}{c} x \, dx$$

$$= \frac{2}{c} \left\{ (c-x) \left(\frac{-\cos n\pi/c \, x}{n\pi/c} \right) \Big|_0^c - (-1) \left(\frac{-\sin n\pi/c \, x}{(n\pi/c)^2} \right) \Big|_0^c \right\}$$

$$= \frac{2}{c} \left\{ \frac{-c}{n\pi} \left(0 - c(1) \right) - \frac{c^2}{n^2\pi^2} (0) \right\}$$

$$= \frac{2}{c} \cdot \frac{c^2}{n\pi} = \underline{\underline{\frac{2c}{n\pi}}} \longrightarrow 01M$$

$$f(x) = \sum_{n=1}^{\infty} \frac{2c}{n\pi} \sin \frac{n\pi}{c} x \longrightarrow 01M$$

MCQ answers.

1. 6 option 2
2. 3/2
3. 3 & 2 are the eigenvalues of A
4. -1
5. $f(x)$ is neither even nor odd.
6. $\begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix}$
7. 1/5
8. $\sqrt{\frac{2}{\pi}} \cdot s \left(\frac{5}{s^2+9} + \frac{2}{s^2+25} \right)$
9. 1/2
10. $Q^T Q = I$.